

Unit 2 Question Packet

Matter & Energy

Name **KEY**

Period

SKILLS

- | | |
|---|--|
| 1. classifying types of MATTER | 4. separating Mixtures |
| 2. interpreting PARTICLE DIAGRAMS | 5. CONVERTING TEMPERATURES |
| 3. identifying PHYSICAL & CHEMICAL PROPERTIES of matter | 6. identifying PHYSICAL & CHEMICAL CHANGES in matter |

SKILL #1: **Classifying types of MATTER** - refer to your notes and RB p. 1-2

Classify each of the following with the combination of terms listed below.

pure substance – element

pure substance – compound

mixture – homogeneous

mixture – heterogeneous

- | | |
|--|---|
| 1. HCl (aq) <u>mixture – homogeneous</u> | 2. sugar <u>pure substance – compound</u> |
| 3. KBr (s) <u>pure substance – compound</u> | 4. soil <u>mixture – heterogeneous</u> |
| 5. Cl ₂ (g) <u>pure substance – element</u> | 6. water <u>pure substance – compound</u> |
| 7. CH ₂ (OH) ₂ (aq) <u>mixture – homogeneous</u> | 8. sodium <u>pure substance – element</u> |
| 9. Hg (ℓ) <u>pure substance – element</u> | 10. iron oxide <u>pure substance – compound</u> |
| 11. NH ₃ (ℓ) <u>pure substance – compound</u> | 12. salt water <u>mixture – homogeneous</u> |

13. Matter that is composed of two or more different elements chemically combined in a fixed proportion is classified as

(1) a compound (2) an element (3) a mixture (4) a solution

14. A compound differs from an element in that a compound

(1) is homogeneous (3) has a definite composition
(2) has a definite melting point (4) can be decomposed by a chemical reaction

15. A compound differs from a mixture in that a compound always has a

(1) homogeneous composition (3) minimum of three elements
(2) maximum of two elements (4) heterogeneous composition

16. A heterogeneous material may be

(1) an element (2) a compound (3) a pure substance (4) a mixture

17. Which statement is an identifying characteristic of a mixture?

(1) a mixture can consist of a single element
(2) a mixture can be separated by physical means
(3) a mixture must have a definite composition by weight
(4) a mixture must be homogeneous

18. Which must be a mixture of substances?

(1) solid (2) liquid (3) gas (4) solution

19. Which substance can be decomposed by chemical means?

- (1) aluminum (2) octane (3) silicon (4) xenon

20. Which substance can be decomposed by chemical means?

- (1) ammonia (2) oxygen (3) phosphorus (4) silicon

21. Which substance can not be broken down by a chemical reaction?

- (1) ammonia (2) argon (3) methane (4) water

22. Two substances, A and Z, are to be identified. Substance A can not be broken down by a chemical change. Substance Z can be broken down by a chemical change. What can be concluded about these substances?

- (1) Both substances are elements.
(2) Both substances are compounds.
(3) Substance A is an element and substance Z is a compound.
(4) Substance A is a compound and substance Z is an element.

23. Tetrachloromethane, CCl_4 , is classified as a

- (1) compound because the atoms of the elements are combined in a fixed proportion
(2) compound because the atoms of the elements are combined in a proportion that varies
(3) mixture because the atoms of the elements are combined in a fixed proportion
(4) mixture because the atoms of the elements are combined in a proportion that varies

24. Two different samples decompose when heated. Only one of the samples is soluble in water. Based on this information, these two samples are

- (1) both the same element (2) two different elements
(3) both the same compound (4) two different compounds

25. The table below shows the mass and volume data for four samples of substances at the same temperature and pressure.

The table below shows mass and volume data for four samples of substances at 298 K and 1 atmosphere.

Masses and Volumes of Four Samples

Sample	Mass (g)	Volume (mL)
A	30.	60.
B	40.	50.
C	45	90.
D	90.	120.

Which two samples could consist of the same substance?

- (1) A and B (3) B and C
(2) A and C (4) C and D

Which two samples could consist of the same substance?

_____ and _____

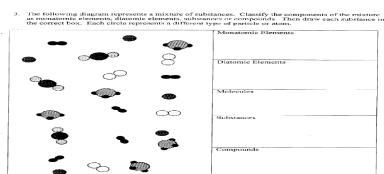
A and C

26. Which terms are used to identify pure substances?

- (1) an element and a mixture
(2) an element and a compound
(3) a solution and a mixture
(4) a solution and a compound

SKILL #2: Interpreting PARTICLE DIAGRAMS - refer to your notes and RB p. 1-3

27. The following diagram represents a mixture of substances. Each circle represents a different type of particle or atom. Classify the components of the mixture as monatomic elements, diatomic elements, or compounds.



a. monatomic element

d. diatomic element

b. diatomic element

e. compound

c. compound

28. Look at the following particle diagrams. Identify each as a pure substance or a mixture. Then identify its composition as elements only, compounds only, or elements and compounds.

a. Pure Substance or <u>Mixture</u> Elements only Compounds only <u>Elements and Compounds</u>	b. <u>Pure Substance</u> or Mixture <u>Elements only</u> Compounds only Elements and Compounds
c. Pure Substance or <u>Mixture</u> <u>Elements only</u> Compounds only Elements and Compounds	d. <u>Pure Substance</u> or Mixture Elements only <u>Compounds only</u> Elements and Compounds

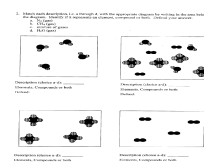
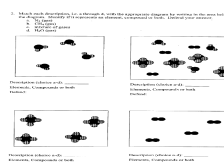
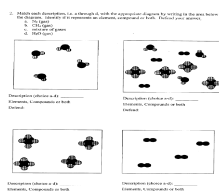
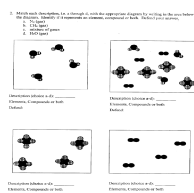
29. Match each description, a through d, with the appropriate diagram. Then identify if it represents an element, compound, or both.

a. N_2

b. CH_4

c. mixture of gases

d. H_2O



d
element
compound

c
element
compound

b
element
compound

a
element
compound

both

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Gas (g)

1. Which substance represents a liquid? Defend your answer.

2. Which substance represents a gas? _____
Defend your answer.

Fig. 1. *Staphylococcus aureus* strains and their classification by phage typing.

Defend your answer.

4. Which substance(s) I listed gave positive

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5. Which substance(s) will take the shape of the container? Defend your answer.

Definite shape	s		Takes the shape of its container	l, g
Definite volume	s, l		Compressable	g
Particles in a fixed/rigid geometric pattern	s		Expands to fill its container	g
Most dense	S		Least dense	g

SKILL #3: Identifying PHYSICAL & CHEMICAL PROPERTIES of matter - refer to your notes and RB p. 13

31. Identify the following as either a physical property (P) or a chemical property (C). (*Hint: there are only 5 chemical properties listed below, the rest are physical properties.*)

- a. ____P Magnesium is malleable.
- b. ____P Magnesium conducts electricity.
- c. ____C Magnesium reacts with an acid.
- d. ____P Magnesium has a high boiling point.
- e. ____C Iron reacts with oxygen to form rust.
- f. ____C Iron forms a compound with chlorine.
- g. ____P Salt dissolves in water.
- h. ____C Hydrogen gas burns in air.
- i. ____P Sodium chloride forms an aqueous solution.
- j. ____P Hydrogen gas has a density of 0.00009 g/cm³ at STP.
- k. ____P Hydrogen gas has a boiling point of -253°C at standard pressure.
- l. ____C Oxygen can combine with a metal to produce a compound.
- m. ____P Oxygen gas is slightly soluble in water.
- n. ____P Oxygen gas can be compressed.
- o. ____P Compound A is a blue color.
- p. ____P Water freezes at 0°C.
- q. ____P Gold can be flattened into sheets.
- r. ____P Iron conducts electricity and heat.
- s. ____P Copper can be drawn into a wire.
- t. ____P Oxygen gas is colorless.

32. A large sample of a solid is crushed into smaller pieces for testing. Which two physical properties are the same for both the large sample and one of the smaller pieces?

- (1) mass and density
(3) solubility and density

33. Describe one appropriate laboratory test that can be used to determine the malleability of a solid sample of an element at room temperature. [1]

Place the sample on a solid surface. Strike the sample with a hammer several times to see if the sample flattens.

Try to bend the sample to change the shape.

34. Copper is used in home wiring and electric motors because it has good electrical conductivity. Other uses of copper not related to its electrical conductivity include coins, plumbing, roofing, and cooking pans. Aluminum is also used for cooking pans.

At room temperature, the electrical conductivity of a copper wire is 1.6 times greater than an aluminum wire. The heat conductivity of copper is 1.8 times greater than the heat conductivity of aluminum. The density of copper is 3.3 times greater than the density of aluminum.

- a. Identify one physical property of copper that makes it a good choice for uses that are not related to electrical conductivity. [1]

Copper is very malleable

OR good conductor of heat

OR high melting point

- b. Identify one physical property of aluminum that could make it a better choice than copper for a cooking pan. [1]

An aluminum pan has less mass than a copper pan of the same size because aluminum is less dense

OR Aluminum is less dense than copper

OR A Cu pan would weigh more.

SKILL #4: Separating MIXTURES – refer to your notes and RB p. 2

35. When a mixture of water, sand, and salt is filtered, what passes through the filter paper?

(1) water, only

(3) water and salt

(2) water and sand

(4) water, sand, and salt

36. Recovering the salt from a mixture of salt and water could best be accomplished by

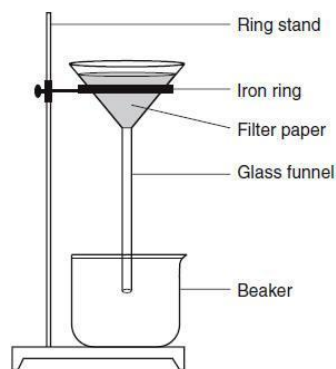
(1) evaporation

(3) paper chromatography

(2) filtration

(4) density determination

37. Which mixture can be separated by using the equipment shown below?



- (1) NaCl(aq) and $\text{SiO}_2(\text{s})$
 (2) NaCl(aq) and $\text{C}_6\text{H}_{12}\text{O}_6(\text{aq})$
 (3) $\text{CO}_2(\text{aq})$ and NaCl(aq)
 (4) $\text{CO}_2(\text{aq})$ and $\text{C}_6\text{H}_{12}\text{O}_6(\text{aq})$

SKILL #5: CONVERTING TEMPERATURES – refer to your notes and RB p. 5

38. Which is not a form of energy?

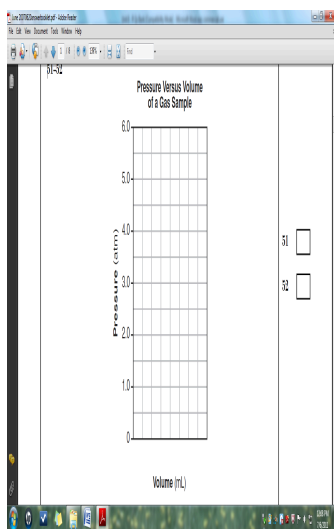
- (1) light (2) temperature (3) electricity (4) heat

39. A student records the temperature of a substance as it is heated using both the Celsius and Kelvin scales. She records the temperatures over time as shown in the table below.

Time (min)	Temperature		Description
	$^{\circ}\text{C}$	K	
0	0	273	ice melts/water freezes
5	20	293	room temperature
10	37	310	body temperature
25	100	373	water boils/steam condenses
35	163	436	oven baking temperature

- a. On the graph, mark appropriate scales on the axes. [1]
 b. Plot the data from the data table. Plot Celsius temperatures using the symbol $\cdot$. Plot Kelvin temperatures using the symbol X. Circle and connect the points, keeping the two lines separate. [1]

Relationship between the Kelvin & Celsius Scales



Temperature
($^{\circ}\text{C}$ and K)

Time (min)

- c. What does the graph reveal about the relationship between the Celsius and Kelvin scales?

The Kelvin scale is always 273 degrees higher than the Celsius scale. If the temperature changes a certain amount on the Celsius scale, it changes the same amount on the Kelvin scale.

40. Convert the following Celsius temperatures into Kelvins – use the equation on Table T and SHOW YOUR WORK: $K = ^\circ C + 273$

a. $5^\circ C + 273 = 278 K$

b. $78^\circ C + 273 = 351 K$

c. $100^\circ C + 273 = 373 K$

d. $250^\circ C + 273 = 523 K$

e. $273^\circ C + 273 = 546 K$

f. $-55^\circ C + 273 = 218 K$

g. $373^\circ C + 273 = 646 K$

h. $400^\circ C + 273 = 673 K$

i. $-10^\circ C + 273 = 263 K$

j. $-157^\circ C + 273 = 116 K$

k. $-273^\circ C + 273 = 0 K$ (Absolute Zero!)

l. $37^\circ C + 273 = 310 K$

m. $25^\circ C + 273 = 298$

41. Convert the following Kelvin temperatures into degrees Celsius – use the equation on Table T and SHOW YOUR WORK: $K = ^\circ C + 273$

a. $5 K - 273 = -268^\circ C$

b. $78 K - 273 = -195^\circ C$

c. $100 K - 273 = -173^\circ C$

d. $250 K - 273 = -23^\circ C$

e. $273 K - 273 = 0^\circ C$

f. $550 K - 273 = 277^\circ C$

g. $373 K - 273 = 100^\circ C$

h. $400 K - 273 = 127^\circ C$

i. $10 K - 273 = -263^\circ C$

j. $157 K - 273 = -116^\circ C$

k. $200 K - 273 = -73^\circ C$

l. $37 K - 273 = -236^\circ C$

m. $25 K - 273 = -248^\circ C$

42. Is it possible to have a negative Kelvin temperature? Explain.

No, it is not possible to have a negative Kelvin temperature because 0 Kelvin is the lowest possible temperature (absolute zero).

43. The temperature of an object changes by $100^\circ C$. The same temperature change in Kelvins would be

(1) $100 K$

(2) $173 K$

(3) $272 K$

(4) $373 K$

44. The difference between the boiling point and the freezing point of pure water at standard pressure is

(1) $32 K$

(2) $273 K$

(3) $100 K$

(4) $373 K$

45. Different masses of copper and iron have the same temperature. Compared to the average kinetic energy of the copper atoms, the average kinetic energy of the iron atoms is

(1) $273 K$ less

(2) $100^\circ C$ less

(3) $273 K$ more

(4) the same

46. The average kinetic energy of water molecules is greatest in which of these samples?

(1) $10 g$ of water at $35^\circ C$

(3) $100 g$ of water at $25^\circ C$

(2) $10 g$ of water at $55^\circ C$

(4) $100 g$ of water at $45^\circ C$

47. As the temperature of a substance decreases, the average kinetic energy of its particles

(1) decreases

(2) increases

(3) remains the same

48. Which change in the temperature of a 1-gram sample of water would cause the greatest increase in the average kinetic energy of its molecules?

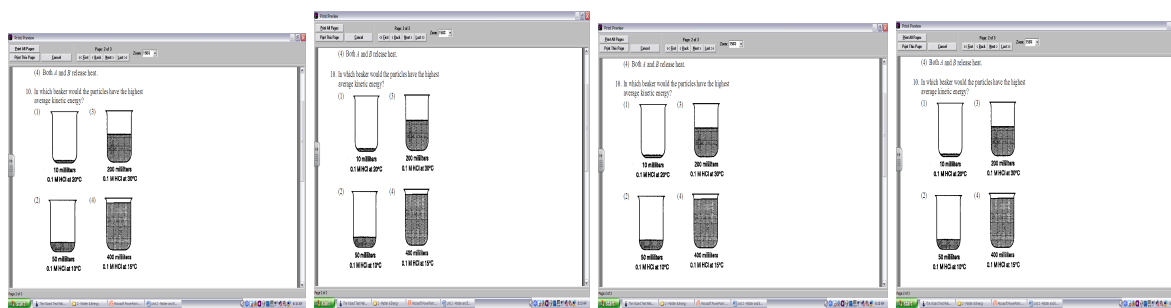
(1) 1°C to 10°C

(2) 10°C to 1°C

(3) 50°C to 60°C

(4) 60°C to 50°C

49. In which beaker would the particles have the highest average kinetic energy? **3**



SKILL #6: Identifying PHYSICAL & CHEMICAL CHANGES in matter – refer to your notes and RB p. 13

50. Identify the following as either a physical change (P) or a chemical change (C). (Hint: there are only 4 chemical changes listed below, the rest are physical changes.)

a. **P** Freezing

b. **P** Boiling

c. **P** Condensing

d. **P** Evaporating

e. **C** Burning

f. **C** Exploding

g. **P** Dissolving

h. **C** Decomposing

i. **P** Forming an aqueous solution

j. **C** Corroding/rusting

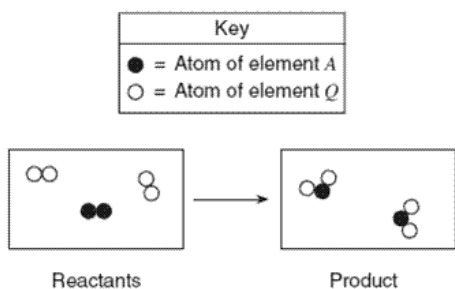
k. **P** Crystallizing

l. **P** Subliming

51. List the three physical changes which are endothermic: **melting, boiling, subliming**

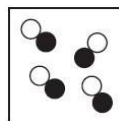
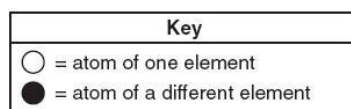
52. List the three physical changes which are exothermic: **condensing, freezing, deposition**

53. The diagram below represents the starting materials (reactants) and ending materials (products) after a change has taken place. Was the change physical or chemical? Explain. [1]

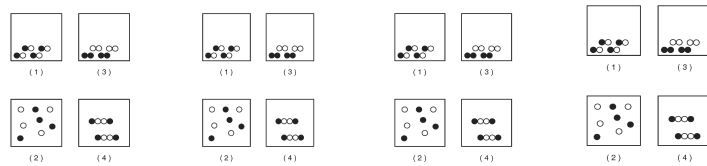


Chemical – a new substance is formed

54. Given the particle diagram representing four molecules of a substance:



Which particle diagram best represents this same substance after a physical change has taken place? **1**



55. In an investigation, a dripless wax candle is massed and then lighted. As the candle burns, a small amount of liquid wax forms near the flame. After 10 minutes, the candle's flame is extinguished and the candle is allowed to cool. The cooled candle is massed.

a. Identify *one* physical change that takes place in this investigation. [1]

Melting, vaporization, solidification

b. State *one* observation that indicates a chemical change has occurred in this investigation. [1]

The burning candle releases heat and light